

A ResNet Approach for AoA and AoD Estimation in Analog Millimeter Wave MIMO Systems

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Abstract—Parametric millimeter-wave (mmWave) channel estimation involves modeling the channel matrix by combining direction-dependent signal paths, exploiting the sparse nature of mmWave channels. In our study, we propose a deep learning-based approach to estimate angle-of-arrival (AoA) and angle-of-departure (AoD) parameters from input observations in the frequency domain. To address this challenge, we have adapted a residual convolutional neural network (ResNet) to this specific problem and incorporated a technique from topological data analysis, enabling us to accurately retrieve the angular frequencies. Furthermore, we have extended this basic architecture by incorporating a posterior model fitting to enhance the system performance in terms of probability of detection. In our research, we compare the ResNet and extended ResNet approaches with state-of-the-art signal processing techniques and the Crámer-Rao lower bound through simulation. Our results indicate significant improvements in system robustness by increasing the probability of detection while maintaining a reduced estimation error.

Index Terms—mmWave, channel estimation, analog beamforming, deep learning, ResNet

I. INTRODUCTION

Communications at millimeter-wave (mmWave) frequencies play a crucial role in 5G technologies and are expected to be also part of future 6G systems [1]. These frequencies present challenges, such as lower penetration capability through obstacles and reduced coverage compared to lower frequencies. To overcome these challenges, advanced, highly directive beamforming techniques are employed to concentrate transmission energy in specific directions. Proper beamforming designs require knowledge on the main outgoing and incoming directions of the communication channel between transmitter (Tx) and receiver (Rx), that is, knowledge of the angle-of-arrival (AoA) and angle-of-departure (AoD) of each path. Besides beamforming design, which becomes a relevant problem in massive multiple-input multiple-output (MIMO) deployments, estimation of AoAs and AoDs is a problem with implications in other areas such as initial alignment [2] or wireless localization and tracking [3].

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In this work, the problem of AoA and AoD estimation is addressed considering a parametric mmWave MIMO system with analog beamforming (ABF). ABF is assumed due to its reduced equipment cost and capability to offer high data rates in single-user environments [4]. Previous works have tackled this problem using signal processing-based algorithms, such as the Transformed Spatial Domain Channel Estimation (TSDCE) [5] or 2D-Discrete Fourier Transforms (DFT) in the frequency domain [6]. Recently, artificial intelligence methods have been applied to several problems in the field of wireless communications [7], including the specific case of channel estimation. For example, the work in [8] explores the use of deep neural networks for mmWave channel estimation in massive MIMO. In the field of tracking AoA/AoD values, and assuming a single source and ABF, we can find solutions based on recurrent neural networks (RNN) [9] or long short-term memory (LSTM) architectures [10]. These diverse approaches highlight the ongoing efforts in both conventional signal processing and emerging machine learning methods within this field.

In this paper, a deep-learning (DL)-based method is proposed, which exploits the sparse nature of the mmWave channel, with the aim of obtaining an adaptive and low-complexity solution. In the literature, there are different neural network architectures that focus on sparse signal recovery for one-dimensional signals [11] and for two-dimensional signals [12]. In our case, the estimation is tackled as a supervised image-to-image translation problem, by updating a residual convolutional neural network (ResNet)-based architecture, known as 2D-DeepFreq [12], originally proposed to estimate the frequencies of multiple two-dimensional sinusoids. Our approach builds upon the system model proposed in [5], where the observation matrix conformed by the array outputs can be interpreted as the DFT of a sum of 2D complex sinusoids defined by the existing channel paths. This matrix is fed to the ResNet with the aim of performing signal denoising and peak enhancement, providing as output a real image that encodes the underlying paths as localized Gaussian kernels. As a result, the task of finding 2D frequencies is analogous to searching for local maxima in a 2D grayscale image. Considering this, the use of a peak search algorithm based on the persistent homology (PH) algorithm [13] is proposed to enhance the ResNet

output. The PH method combines concepts from persistent homology and topological data analysis (TDA) to detect and quantify peaks in data. To further improve the robustness of the proposed architecture, an extended architecture is as well proposed by incorporating a posterior model fitting technique.

II. SYSTEM MODEL

Let us assume a single-user mmWave MIMO geometric channel, where both the (Tx) and (Rx) are equipped with uniform linear arrays of n_t and n_r antenna elements, respectively. L denotes the number of scatterers, each one contributing with a single Tx-Rx propagation path. The complex channel coefficient for each path is defined by α_l , $l = 1, \dots, L$, while ψ_l and ϕ_l stand for the AoA and AoD, respectively. Using the full parametric model, the channel is expressed as:

$$\mathbf{H}(\boldsymbol{\theta}) = \sqrt{n_t n_r} \sum_{l=1}^L \alpha_l \mathbf{a}_r(\psi_l) \mathbf{a}_t^H(\phi_l), \quad (1)$$

where $\boldsymbol{\theta} \triangleq [|\alpha_1|, \angle\alpha_1, \phi_1, \psi_1, \dots, |\alpha_L|, \angle\alpha_L, \phi_L, \psi_L]^T$ is the parameter vector. Note that $|\alpha_l|$ and $\angle\alpha_l$ stand for the magnitude and phase of each channel coefficient. α_l -s are independent identically distributed (i.i.d.) random variables with distribution $\alpha_l \sim \mathcal{CN}(0, \sigma_\alpha^2/L)$. AoAs (ψ_l) and AoDs (ϕ_l) are drawn from a uniform distribution $\in [0, 2\pi]$.

The antenna array responses at the Tx and Rx, under a half-wavelength antenna separation assumption, are

$$\mathbf{a}_t(\phi_l) = \frac{1}{\sqrt{n_t}} [1, e^{-j\pi \cos \phi_l}, \dots, e^{-j\pi(n_t-1) \cos \phi_l}]^T, \quad (2)$$

$$\mathbf{a}_r(\psi_l) = \frac{1}{\sqrt{n_r}} [1, e^{-j\pi \cos \psi_l}, \dots, e^{-j\pi(n_r-1) \cos \psi_l}]^T. \quad (3)$$

The problem of estimating the channel $\mathbf{H}(\boldsymbol{\theta})$ becomes a problem of estimating the parameters in $\boldsymbol{\theta}$. To this end, an open-loop pilot-based training phase is considered. The beam search space is given by a codebook comprising P and Q quantized angles at the Tx and Rx side, respectively, denoted by $\bar{\phi}_p$ for $p = 0, 1, \dots, P-1$, and $\bar{\psi}_q$ for $q = 0, 1, \dots, Q-1$. The codebook angles are selected as in [5] as follows:

$$\cos(\bar{\phi}_p) = \mathcal{W}_{[-1,1]} \left(\frac{2p}{P} \right), \quad \cos(\bar{\psi}_q) = \mathcal{W}_{[-1,1]} \left(-\frac{2q}{Q} \right), \quad (4)$$

where $\mathcal{W}_{[a,b]}(x) \triangleq x - (b-a) \left\lceil \frac{x-b}{b-a} \right\rceil$ is the $[a, b]$ wrapping operator with $\lceil \cdot \rceil$ denoting the ceiling function. The above conditions imply simultaneously a uniform quantization in the range $[-1, 1]$ for the cosine of the codebook angles and a specific codebook ordering at the Tx and the Rx.

Beamforming/combining vectors are calculated to match the channel response [5], so $\mathbf{f} = \mathbf{a}_t(\bar{\phi}_p)$ and $\mathbf{w} = \mathbf{a}_r(\bar{\psi}_q)$. For each $\{q, p\}$ direction pair, the received signal is

$$y_{q,p} = \sqrt{\rho} \mathbf{w}_q^H \mathbf{H} \mathbf{f}_p + \mathbf{w}_q^H \mathbf{n}, \quad (5)$$

where $\rho \in \mathbb{R}^+$ is the transmit power. The noise term $\mathbf{n} \sim \mathcal{CN}(0, \boldsymbol{\Sigma}_n)$ is a complex additive white Gaussian noise $1 \times n_r$ vector with covariance $\boldsymbol{\Sigma}_n = \sigma_n^2 \mathbf{I}_{n_r}$, where $\mathbf{I}_{n_r}$ stands for the $n_r \times n_r$ identity matrix. The symbol s is set to 1 during training

for the sake of simplicity. Hence, the system signal-to-noise ratio (SNR) is given by ρ/σ_n^2 . The observation matrix $\mathbf{Y} \in \mathbb{C}^{Q \times P}$ is formed after transmitting the pilot symbol through all the direction combinations. Considering a specific DFT-based codebook for ABF, it can be shown that $\mathbf{Y}$ corresponds to the DFT of a sum of L complex sinusoids associated to each of the paths making up the channel $\mathbf{H}$, and whose frequencies are directly related to the underlying AoD and AoA [5].

III. RESNET BASED AOA AND AOD ESTIMATION

In this section, the two ResNet-based approaches considered in this paper are presented. The first one is a basic ResNet architecture combined with TDA, while the second one is an extension which incorporates a posterior model fitting technique.

A. 2D-ResNet Architecture

First, we consider a DL-based method adapted from the 2D-DeepFreq model in [12], which is based on a convolutional ResNet-like architecture. A block diagram of the model architecture is shown in Fig. 1. The method aims at addressing the AoA and AoD estimation problem by identifying the complex sinusoids present in the input complex-valued observation matrix $\mathbf{Y}$. In contrast to the 2D-DeepFreq model, our input $\mathbf{Y}$ already encodes a frequency-domain representation of the sum of multiple sinusoids. Therefore, the matched filtering module present in DeepFreq before the up-sampling stage has been suppressed. As shown in Fig. 1, the input $\mathbf{Y}$ is first decomposed into its real and imaginary parts $\mathbf{Y}_{\Re} \in \mathbb{R}^{Q \times P}$ and $\mathbf{Y}_{\Im} \in \mathbb{R}^{Q \times P}$, respectively. These matrices are then modified through an upsampling step (Up 1) which increases their dimensions to $Q' = \beta Q$ and $P' = \beta P$, resulting in new matrices $\tilde{\mathbf{Y}}_{\Re} \in \mathbb{R}^{Q' \times P'}$ and $\tilde{\mathbf{Y}}_{\Im} \in \mathbb{R}^{Q' \times P'}$. We will denote by w_n and h_n the width and height of the output feature map at the n -th layer, by c_n for the number of the output channels at the n -th layer, by f_n for the number of filters at the n -th layer, and by k_n for the length of the filter kernels (assuming they are square). The size of all filters utilized in the convolutions is consistently set to $k_n = k = 5$, as specified in [14]. The upsampling step is performed through a nearest-neighbor interpolation method [15] and a 2D transposed convolution layer (ConvTranspose2D) with a stride of 2:

$$\text{ConvTBlock}_0(x_0) := \text{ConvTranspose2D}_{k \times k}(x_0, \text{stride} = 2),$$

where x_0 denotes the input to the initial layer. Then, the result enters the super-resolution module which iteratively stacks 64 residual blocks (RBs), as described in [14]. As shown in Fig. 1, RBs follow the Conv-BN-ReLU order, where ReLU is the rectified linear unit function and BN stands for batch normalization. For a generic input x to any RB, the output is given by:

$$\text{RB}(x) := \text{ReLU}(g(x) + x),$$

where

$$g(x) := \text{BN}(\text{Conv2D}_{k \times k}(\text{ReLU}(\text{BN}(\text{Conv2D}_{k \times k}(x)))).$$

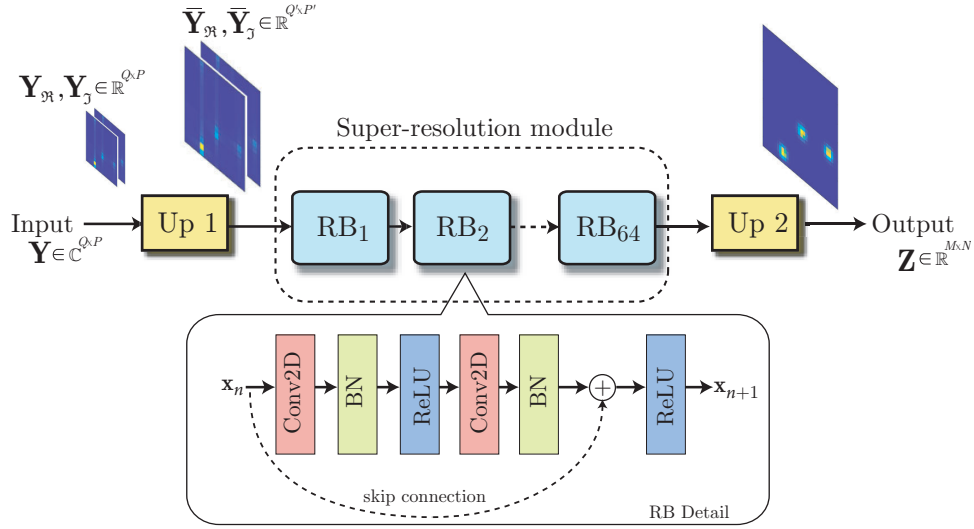


Fig. 1. 2D-ResNet model architecture.

Finally, another ConvTranspose2D with a stride of 2 is applied for a second upsampling (Up 2). The output of the network is a matrix $\mathbf{Z} \in \mathbb{R}^{M \times N}$, where M and N are selected as integer multiples of the input dimensions Q' and P' . In this work we set $M = N = 256$ and the minimum upsampling factor $\beta = 2$.

The considered ground truth target corresponds to a high-resolution matrix $\mathbf{X} \in \mathbb{R}^{M \times N}$ built as the sum of L 2D-Gaussian functions centered in the true AoA and AoD for each path [12]. The elements of matrix $\mathbf{X}$, for $m = 0, \dots, M - 1$ and $n = 0, \dots, N - 1$, are constructed as follows:

$$\mathbf{X}_{m,n} = \frac{1}{2\pi\sigma_x\sigma_y} \sum_{l=1}^L e^{\left(\frac{(\omega_m - \tilde{\omega}_{\psi_l})^2}{2\sigma_x^2} + \frac{(\omega_n - \tilde{\omega}_{\phi_l})^2}{2\sigma_y^2} \right)}, \quad (6)$$

where $\omega_m = m \frac{2\pi}{M}$, $\omega_n = n \frac{2\pi}{N}$ and $\tilde{\omega}_{\phi_l} = \mathcal{W}_{2\pi}(\omega_{\phi_l})$, $\tilde{\omega}_{\psi_l} = \mathcal{W}_{2\pi}(\omega_{\psi_l})$ are the 2π -wrapped ground-truth frequencies. The parameters σ_x and σ_y represent the standard deviations that define the width of the Gaussian functions.

The loss function corresponds to the mean square error (MSE) between the ground truth and the network output:

$$\text{Loss} = \frac{1}{MN} \sum_{n=0}^{N-1} \sum_{m=0}^{M-1} (\mathbf{Z}_{m,n} - \mathbf{X}_{m,n})^2. \quad (7)$$

The ResNet output matrix $\mathbf{Z}$ is interpreted as a denoised observation matrix containing a set of local maxima that identifies the mmWave channel paths. Thus, a peak search algorithm is used to find the corresponding AoAs and AoDs in the test stage. It is worth noting that the task of finding 2D frequencies is analogous to searching for local maxima in a 2D grayscale image. Therefore, we consider a peak search algorithm based on the PH algorithm [13], which makes use of concepts from TDA to detect and quantify peaks in data. The PH algorithm has applications in multiple fields ranging from image analysis to sensor data processing, where it is able

to uncover hidden structures and patterns in the data that may not be apparent through traditional data analysis methods. The result of applying the PH algorithm produces two vectors $\tilde{\omega}_{\phi'_l}$ and $\tilde{\omega}_{\psi'_l}$, for $l = 1, 2, \dots, L$ which are the locations of the peaks. In order to recover the values of ϕ'_l and ψ'_l we have to undo the angle-to-frequency mapping as in Eq.(12) in [5]. The stages comprising the above procedure are summarized in Algorithm 1, except for the final “model fitting” stage, which is a component of the extended model described in the following section.

Algorithm 1: 2D-ResNet for Frequency Estimation

Model Training

Data: Complex data samples $(\mathbf{Y}_{\mathcal{R}}, \mathbf{Y}_{\mathcal{I}})$, ground truth path matrix $\mathbf{X}$

Generate Dataset: $\mathcal{D} \leftarrow \{(\mathbf{Y}_{\mathcal{R}}, \mathbf{Y}_{\mathcal{I}}), \mathbf{X}\}_{i=1}^{N_{data}}$.

Train ResNet: $\text{ResNet} \leftarrow \text{TrainResNet}(\mathcal{D})$

Result: Trained model **ResNet**

Angle Recovery

Predict: $\mathbf{Z} \leftarrow \text{ResNet}(\mathbf{Y}_{\mathcal{R}}, \mathbf{Y}_{\mathcal{I}})$

Output: Angles $\{(\tilde{\omega}_{\phi'_l}, \tilde{\omega}_{\psi'_l})\}_{l=1}^L \leftarrow \text{PeakDetection}(\mathbf{Z})$

Model Fitting

Fitting:

$\{(\tilde{\omega}_{\phi''_l}, \tilde{\omega}_{\psi''_l})\}_{l=1}^L \leftarrow \text{Fitting}(\mathbf{Z}, \{(\tilde{\omega}_{\phi'_l}, \tilde{\omega}_{\psi'_l})\}_{l=1}^L)$

Output: Fitted peak parameters $\{(\phi''_l, \psi''_l)\}_{l=1}^L$

B. Extended 2D-ResNet with Model Fitting

Peak detection methods such as PH often face challenges related to the resolution grid, which can impact their reliability.

One way to improve the peak detection is to refine its results using model fitting techniques. In this section, we consider a model fitting method to enhance the accuracy of the output after using the PH technique, leading to an improved AoA/AoD estimation scheme.

To improve the accuracy of the estimated angles without being restricted by the grid resolution, we apply a least-squares model fitting procedure. To this end, we fit one after another, a displaced 2D-Gaussian kernel ($\mathcal{G}_\mu$) over the ResNet output $\mathbf{Z}$, initialized each time on one of the locations corresponding to the estimated angle pair $(\tilde{\omega}_{\psi_l}, \tilde{\omega}_{\phi_l})$. Then, for each of the paths, we solve a minimization problem as follows:

$$\min_{\mu} \sum_{n=0}^{N-1} \sum_{m=0}^{M-1} (\mathcal{G}_\mu(m, n) - \mathbf{Z}_{m,n})^2, \quad (8)$$

where $\mu = (\tilde{\omega}_{\psi_l}, \tilde{\omega}_{\phi_l})$ and the displaced Gaussian kernel is

$$\mathcal{G}_\mu(m, n) = Ae \left(\frac{(\omega_m - \tilde{\omega}_{\psi_l})^2}{2\sigma_x^2} + \frac{(\omega_n - \tilde{\omega}_{\phi_l})^2}{2\sigma_y^2} \right). \quad (9)$$

The minimization starts by using as initial guess $\mu_0 = (\tilde{\omega}_{\psi_l}, \tilde{\omega}_{\phi_l})$. In this work, we adopt the Levenberg-Marquardt algorithm [16] to solve the above non-linear minimization problem. The obtained location can be again mapped to the final estimated angle pair (ϕ_l'', ψ_l'') .

C. Complexity

The computational complexity of the ResNet convolutional neural network architecture can be obtained by analyzing its different stages. As shown in [12], the complexity of the n -th convolutional layer can be calculated as $h_n w_n c_{n-1} k_n^2 f_n$. After considering all the convolutional layers and performing an asymptotic analysis, it can be observed that the method involves an overall complexity order $\mathcal{O}(PQ)$. Additionally, the PH algorithm on the sublevel set filtration of a real-valued signal has a complexity of $\mathcal{O}(m \log(m))$, where m represents the size of the input signal, which in our case results in $\mathcal{O}(PQ \log(PQ))$. Then, the complexity of the ResNet is dominated by the term related to the PH algorithm.

The extended ResNet further includes the complexity of the model fitting stage. The complexity order of the Levenberg-Marquardt algorithm for curve fitting in a two-dimensional Gaussian function can be expressed as $\mathcal{O}(k(mp + mp^2))$, where k is the maximum number of iterations until convergence, m is the number of data points and p is the number of parameters to optimize. We perform the Levenberg-Marquardt algorithm for $p = 2$ (i.e. to optimize μ_x and μ_y) independently for L non-overlapping Gaussian functions and $m = P'Q'$, leading to a total complexity order $\mathcal{O}(PQ)$. The maximum number of iterations is set to 1200 as stopping criterion, although the actual number is much lower. In this case, the extended ResNet complexity is as well dominated by the term related to the PH algorithm, i.e., $\mathcal{O}(PQ \log(PQ))$.

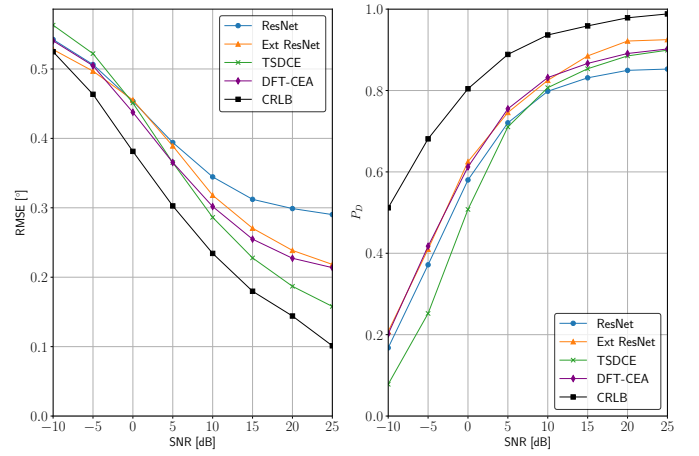


Fig. 2. RMSE and P_D of channel estimation for different values of SNR, with $L = 3$, $n_t = n_r = 16$ and $Q = P = 16$.

IV. NUMERICAL RESULTS

The performances of the two proposed ResNet neural network schemes assuming $M = N = 256$ have been compared with two signal processing-based channel estimation methods, namely, the TSDCE [5] and DFT-CEA [6]. The Crámer-Rao lower bound (CRLB) of channel estimation performance considering it as a 2D-frequency estimation problem [5] is also included within the comparison. The training process of the ResNet models was carried out as follows. A dataset composed of random channel realizations for a range of SNRs was constructed, considering a random number of paths from $L = 1$ to 10. The selected optimizer was RMSprop with a learning rate of 0.003 and a batch size of 32. The model was trained for 500 epochs, using a training dataset comprising 10000 input samples (channel realizations), along with a validation dataset of 1000 samples.

To consider an ABF architecture, all the methods are evaluated with a single RF chain. Moreover, all methods are tested under the same pilot overhead, which is directly $Q \times P$ pilots. The number of desired paths is set to L for all methods. The simulations consider 10000 random channel realizations for each SNR value, with $\rho = 1$, i.e. the SNR definition is $1/\sigma_n^2$. Channel coefficients are drawn from a zero mean complex Gaussian distribution with variance $1/L$, while AoA and AoD angles are retrieved from a uniform distribution in the range $[0, \pi]$. For the DFT-CEA, the DFT operation is performed with $N_{DFT} = 1024$ points.

The AoA/AoD estimation accuracy is assessed through the root mean square error (RMSE) of the angle measurements, calculated as

$$\text{RMSE} = \sqrt{\frac{1}{|\mathcal{N}|} \sum_{i \in \mathcal{N}} (\vartheta_i - \hat{\vartheta}_i)^2}, \quad (10)$$

where $\mathcal{N}$ denotes the set of successfully detected angles and ϑ_i considers any AoA or AoD. We also evaluate the probability

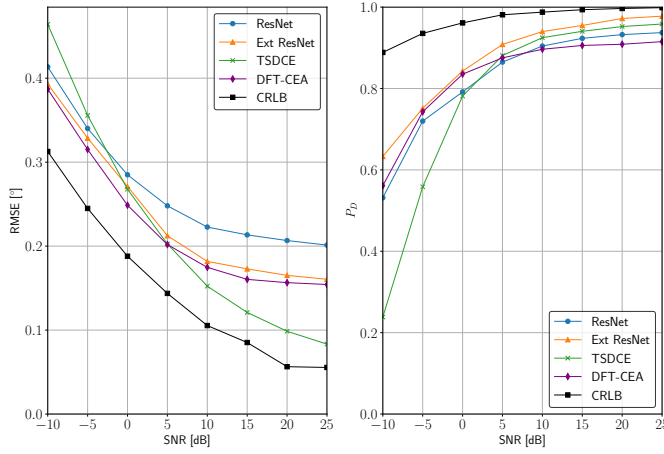


Fig. 3. RMSE and P_D of channel estimation for different values of SNR, with $L = 3$, $n_t = n_r = 32$ and $Q = P = 32$.

of detection (P_D), considering a detection as successful if the associated $\text{RMSE} \leq 1^\circ$.

Numerical results have evaluated, on the one hand, the impact of increasing the antenna array and codebook size, and, on the other hand, the performance of the algorithms when the number of paths L increases. Fig. 2 shows the RMSE and P_D of the estimated AoA/AoD in channel realizations with $L = 3$ paths, considering a system with $n_t = n_r = 16$ antennas and using a codebook with $Q = P = 16$ directions. The extended ResNet architecture outperforms the ResNet both in RMSE and P_D . At very low SNRs, the extended ResNet comes close to achieving the CRLB. Regarding the comparison with the signal processing-based algorithms, we can see how the ResNet models are competitive in RMSE for low SNRs, while signal processing-based methods show an improved performance for high SNR, and only the extended ResNet is able to remain close to the DFT-CEA. When evaluating the probability of detection (P_D) results, it is observed that the extended ResNet outperforms or equals all other methods, establishing itself as the method to maximize this parameter especially at high SNRs. The joint analysis of RMSE and P_D results for the ResNet approaches suggests that these schemes are able to detect a larger set of AoA/AoD angles (including those close to the 1° threshold), which leads to larger RMSEs, due to this metric averaging over all the detected angles. This situation bears significant relevance in the application scenario, as it suggests that while signal processing baselines excel in high SNR conditions, the extended ResNet excels when detection reliability is critical across various SNR levels.

Next, we evaluate in Fig. 3 the impact of increasing the array and codebook size to $n_t = n_r = 32$ and $Q = P = 32$. A consistent performance trend is evident across the algorithms, characterized by reduced errors and improved probability of detection. Results in this figure again suggest that the extended ResNet is capable of detecting a larger set of angles, including the ones close to the established threshold, while signal processing-based methods achieve a reduced RMSE

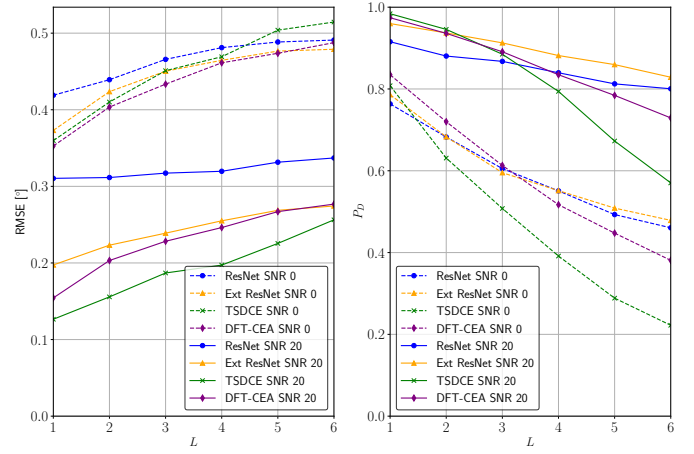


Fig. 4. RMSE and P_D of channel estimation for different values of L , with SNR = 0 and SNR = 20, $n_t = n_r = 16$ and $Q = P = 16$.

but detect less angles. Particularly noteworthy is the extended ResNet's sustained superiority in terms of P_D , remaining competitive to the DFT-CEA approach in RMSE within a reduced complexity. Note that the extended ResNet complexity is dominated by the term $\mathcal{O}(PQ \log(PQ))$, while TSDCE complexity is given by $\mathcal{O}((L-1)PQ \min(P, Q))$ and DFT-CEA complexity is $\mathcal{O}(N_{DFT}^2 \log_2 N_{DFT})$ [5]. For $P = Q$, the extended ResNet complexity boils down to $\mathcal{O}(P^2 \log(P))$, showing less complexity than TSDCE and DFT-CEA (note that, in general, $\{P, Q\} < N_{DFT}$).

The effect of increasing the number of paths can be seen in Fig. 4, which shows the results of ranging parameter L for two SNR values and $M = N = 256$. As it can be seen, in terms of RMSE, for both SNR values, the extended ResNet is the method with the closest performance to the DFT-CEA, while at high SNR the TSDCE is still the approach providing the most accurate angle estimation for the considered range of L . Regarding the P_D , both ResNet approaches are able to detect more angles as the number of paths increases, showing a larger robustness than the signal processing-based methods. Indeed, their curves show a smaller slope than the one of the DFT and TSDCE methods. This versatility is very valuable in real-world applications where wireless communication faces challenging signal environments, ranging from weak and noisy conditions to more favorable ones.

V. CONCLUSION

In this study, we introduced and evaluated two distinct adaptations of a residual convolutional neural network (ResNet) to tackle the challenging task of estimating the angle-of-arrivals (AoAs) and angle-of-departures (AoDs) in a parametric mmWave channel system formulation. This estimation was framed as a supervised image-to-image translation problem within the frequency domain. We observed that ResNet architectures are able to detect a larger set of AoA/AoD angles, leading to larger average mean square errors than signal processing-based benchmarks. The extended ResNet architec-

ture achieves a superior probability of detection, even when increasing the number of paths, while remaining competitive in average mean square error with respect to signal processing-based algorithms, within a reduced computational complexity. Our findings support the adaptability and potential of deep learning approaches for solving physical layer tasks such as AoA/AoD channel estimation, and highlights the significance of considering signal conditions when designing competitive solutions.

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